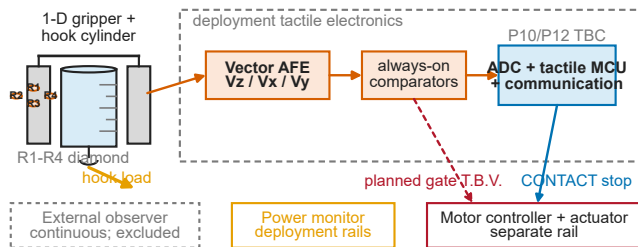
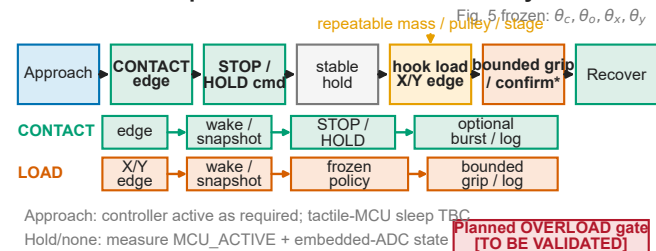


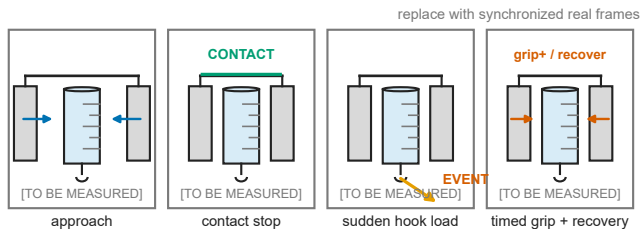
a System, deployment boundary and instruments



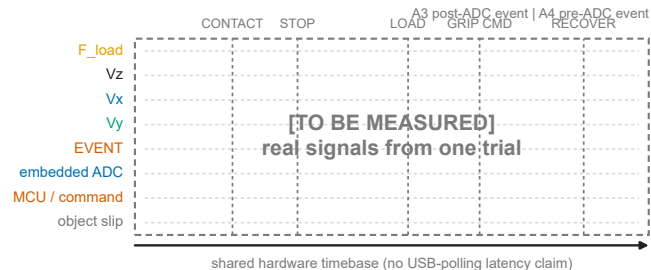
b Frozen task protocol with two GPIO event cycles



c Same-trial action frames: hook-load reflex

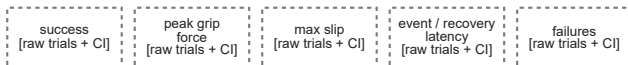
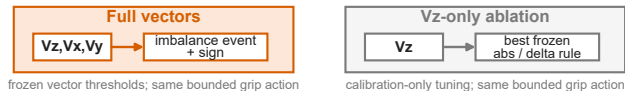


d Same-clock disturbance-event-action data slots



e Causal ablation and task-level completion metrics

A4, identical task and frozen thresholds

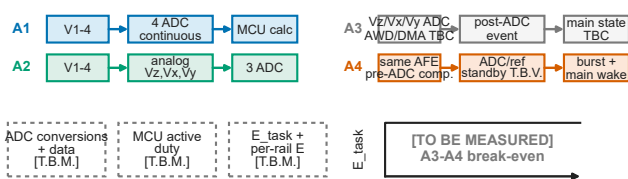


[TO BE MEASURED] retain safety stops and unrecovered trials; fixed hook direction does not prove signed control

f A1-A4 resource, energy and break-even audit

vectorization: A1 vs A2

event location: A3 vs A4



actuator energy measured and reported separately

event rate / idle duty
A3 may gate main-MCU duty + communication
A4 extra = avoid idle quantization; standby to be verified